

Explanation

Measured values

U	Voltage
f	Frequency
I _{Av}	Average current

Cos(φ)	Power factor
n	Speed

T _{media}	Water temperature
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Q _m	Measured flow
H _m	Measured Total Head
P _{1m}	Measured Motor Power Input

Calculated values

Q(n)	Flow at nominal speed
H(n)	Total Head at nominal speed
P ₁ (n)	Motor Power Input at nominal speed

η _{total}	Total Efficiency
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Formulas

Q(n)	$Q_m \times (n_{nom}/n)$
H(n)	$H_m \times (n_{nom}/n)^2$
P ₁ (n)	$P_{1m} \times (n_{nom}/n)^3$
η _{total}	$(P \times Q_m \times H_m \times g) / P_{1m}$

H	Head_Sta + Head_Dyn + Head_Geo + Head_J
Head_Static	Static Pressure head
Head_Dyn	Dynamic head
Head_Geo	Geometric elevation head
Head_J	Friction head
Head_C	Friction loss of connector

Legend and test conditions

- Measurements were made with airless water at approximately 20 °C (63 °F) and a kinematic viscosity of 1 mm²/s (= 1.076 × 10⁻⁵ ft²/s = 1 cSt).
- The testbed is constructed and calibrated according to requirements in ISO 9906.

Test Facility

Company	Grundfos Manufacturing Hungary Ltd.
Address	Holland fasor 15.
City	Székesfehérvár
ZIP Code	H8000
Country	Hungary
Phone	Phone:(+36) 22 801 801
Website	www.grundfos.com

Test Report – Performance curve

ISO 9906:2012 Grade 3B2

System Identification

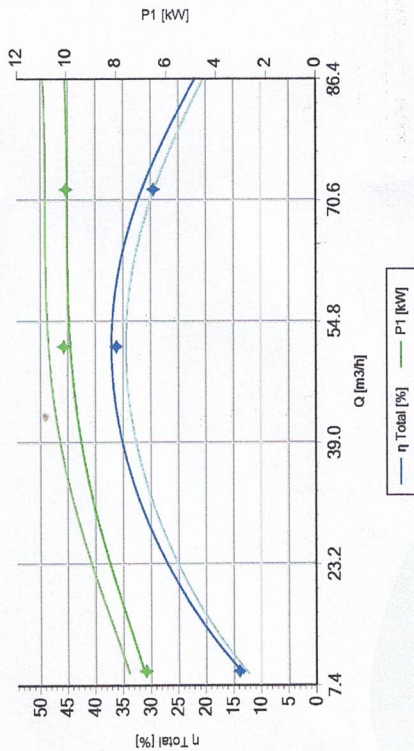
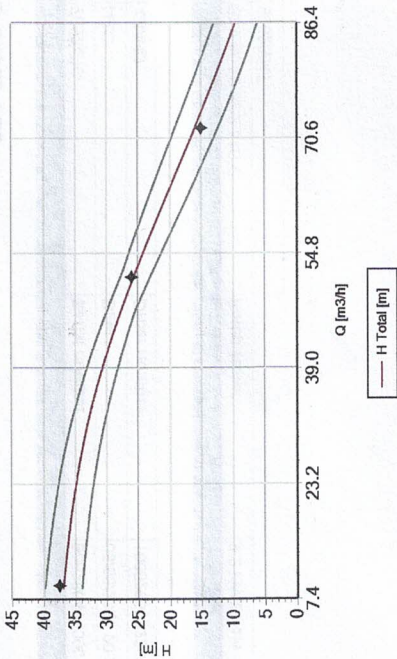
Testbed	76S24PC	Serial number	9862604710001770
Operator	NRC	Date	05/06/2026 20:23
Customer	GM-HD	Order number	100020503236

Pump Data

Pump type	SLV.80.80.92.2.51D.C	Pump Motor	9.2 kW/2 Pole
Product number	98626047		

Measured values for tested pump

The marks indicate the measured values. The tolerance limits are indicated by grey curves.



Result	Qm [m³/h]	Hm [m]	η _{total} [%]	t _{media} [°C]
Point 1	72.00	15.09	29.4	24.8
Point 2	51.54	26.06	36.1	24.8
Point 3	9.29	37.53	13.9	24.8

Result	U1 [V]	U2 [V]	U3 [V]	f [Hz]	I _{Avg} [A]	I1 [A]	I2 [A]	I3 [A]	P1m [kW]
Point 1	398.0	399.0	400.0	50	17.20	17.41	16.64	17.62	10.02
Point 2	400.0	400.0	401.0	50	17.33	17.97	16.76	17.42	10.11
Point 3	398.0	398.0	400.0	50	12.79	13.25	12.30	12.92	6.83

Head components	Head_Static [m]	Head_Dyn [m]	Head_Geo [m]	Head_J [m]
Point 1	15.3	0.73	-0.94	0.03
Point 2	26.6	0.37	-0.94	0.01
Point 3	38.5	0.01	-0.94	0.00

Additionally, during manufacturing the following tests have been carried out:

Tightness of the motor: Leakage Test
High voltage test according to EN 60034-1.

No-load test.
Insulation resistance test.
Earth continuity test.

Performance test description:

This pump is tested according to ISO 9906:2012 grade 3B or ANSI/HI 11.6:2012 grade 3B tolerances. The testbed is constructed and calibrated according to requirements in ISO 9906. Pump is approved according to the tolerances for entire curve, specified in grade 3B.